How to Run the Code for Crude Oil Price Forecasting

Step 1: Set up a Virtual Environment

It is highly recommended to use a virtual environment.

1. Navigate to your project folder.

2. Create a virtual environment:

   python -m venv venv

3. Activate the environment:

   - On Windows: venv\Scripts\activate

   - On macOS/Linux: source venv/bin/activate

Step 2: Install Dependencies

Run the following command inside the activated environment:

   pip install -r requirements.txt

Step 3: Required Data File

Place the following file in the same directory as the code:

   - CMO-Historical-Data-Monthly.xlsx

This file contains the monthly commodity price data required for model training and evaluation.

Step 4: Run the Code

You can execute the full pipeline using either of the provided files:

Option A: Run the Jupyter Notebook (recommended for step-by-step inspection)

```
jupyter notebook code_full.ipynb
```

Option B: Run as a Python script (batch execution)

```
python code_full.py
```

Make sure the dataset and all required packages are accessible to the script or notebook.